

CLASS 5 — WEEK 2

Metabolism, Part 1

How your cells harvest energy from food

Dr. Giorgio Lagna • Monday, April 20, 2026

BIOL 40C • Human A&P III • Foothill College

From Digestion to Metabolism

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CONCEPT

Last week: DIGESTION

We followed food from mouth to colon.

By the time nutrients crossed the intestinal wall, they became:

- Glucose (from carbohydrates)
- Fatty acids + glycerol (from fats)
- Amino acids (from proteins)

These are the building blocks. Now what?

This week: METABOLISM

What happens after absorption?

Building blocks go two ways:

CATABOLISM → break them down to make ATP

ANABOLISM → use them to BUILD your body

Today = catabolism. Wednesday = anabolism + regulation.

By the end of today, you should be able to...

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CONCEPT

- 1** **DISTINGUISH** catabolism from anabolism, and identify which one dominates in the fed vs. fasted state.
- 2** **DESCRIBE** the three stages of glucose catabolism: glycolysis, citric acid cycle, and oxidative phosphorylation.
- 3** **EXPLAIN** how the electron transport chain uses NADH and FADH₂ to pump H⁺ and drive ATP synthase.
- 4** **COMPARE** how fatty acids and amino acids feed into the same mitochondrial machinery as glucose.
- 5** **APPLY** these pathways to a patient case: why does metformin work? Why does cyanide kill?

Before we start: Quick KC Quiz 2 debrief

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CHECK

Most-missed questions — quick clarifications

Q3 NADH → ETC. NADH is a mobile electron carrier; its job is to drop electrons into the ETC, not to directly make ATP.

Q7 ATP synthase = rotor, not enzyme choice. The proton gradient spins it; spinning phosphorylates ADP. Think mini turbine.

Q9 Amino acids can enter the citric acid cycle. When glucose is low, your body will burn protein. We'll see why this is a problem.

Module-numbering heads-up: *Your KC Quiz modules 23.3–23.7 match Amerman's 23.4–23.8. Same content, different label.*

MODULE 23.1

Overview of Metabolism

Two directions, one cellular economy

What IS metabolism?

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CONCEPT

Metabolism = the sum of ALL chemical reactions in the body.

CATABOLISM

Breaking down

Large molecules → small molecules
RELEASES energy → captured as ATP

Examples:

- Glucose → $\text{CO}_2 + \text{H}_2\text{O} + \text{ATP}$
- Triglyceride → fatty acids → ATP
- Protein → amino acids → ATP

Dominates when: FASTED / fed but hungry cells need fuel.

ANABOLISM

Building up

Small molecules → large molecules
REQUIRES energy → spends ATP

Examples:

- Glucose → glycogen (storage)
- Fatty acids → triglyceride (storage)
- Amino acids → proteins (muscle!)

Dominates when: FED / right after you eat.

Why ATP? The cellular currency

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CONCEPT

ATP = Adenosine Triphosphate

Think of ATP as a **rechargeable battery**.

ATP → **ADP + P_i** releases energy that powers muscle contraction, ion pumps, nerve firing, DNA replication, EVERY cellular process.

ADP + P_i → **ATP** re-stores energy. This recycling happens ~**your body-weight in ATP per day**.

Without ATP, you die in seconds. That's why metabolism matters.

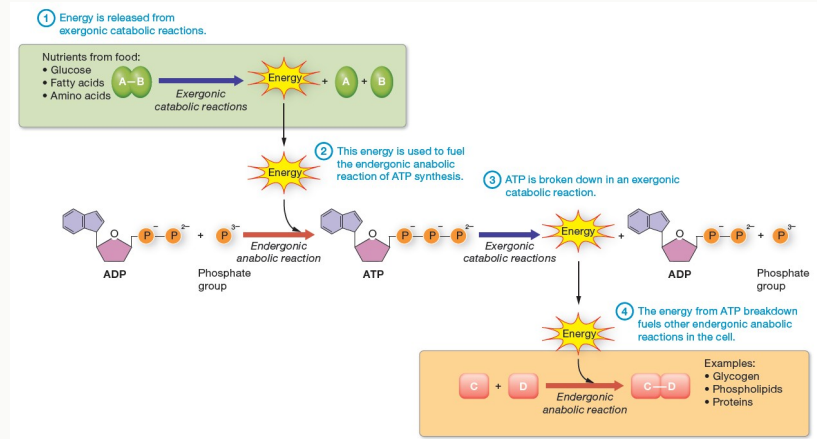


Fig. 23.2 — Catabolism → ATP → anabolism

MODULE 23.2

Glucose Catabolism

Three stages, one molecular machine

The Big Picture: One glucose $\rightarrow$ $\sim$ 30 ATP

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CONCEPT

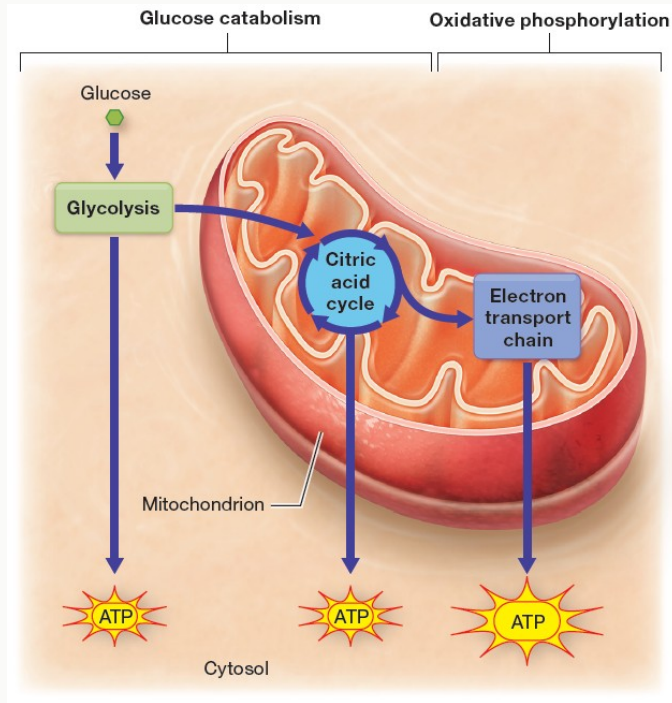


Fig. 23.4 — The 3-stage overview

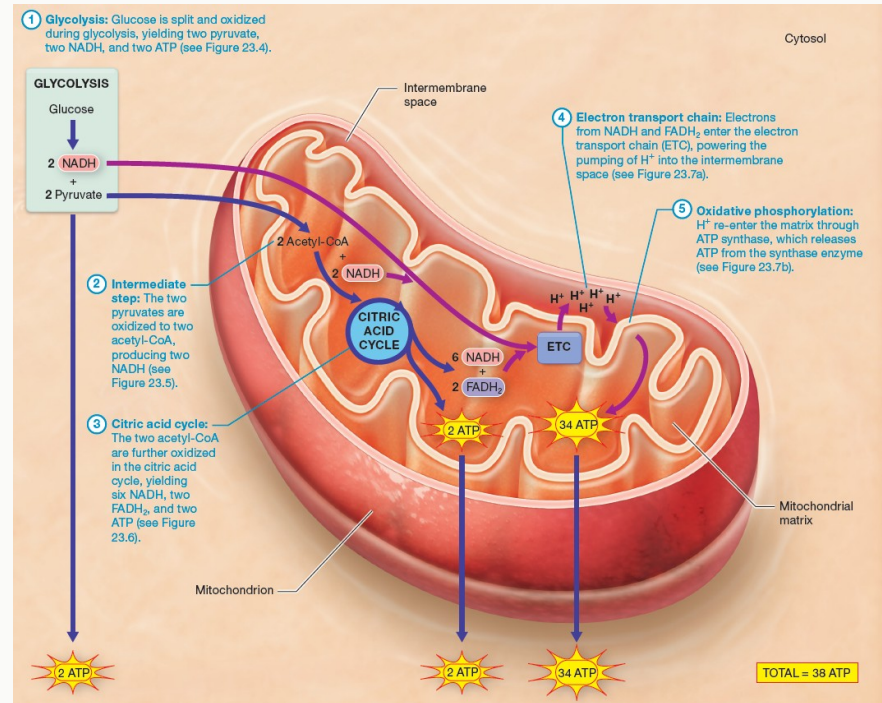


Fig. 23.5 — Same pipeline, more detail. NADH/FADH₂ shuttle electrons to oxidative phosphorylation.

Stage 1 — Glycolysis

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CONCEPT

Glucose → 2 Pyruvate

Where: cytosol (not in mitochondrion)

Oxygen needed? NO (anaerobic)

Net yield per glucose:

- 2 ATP (from substrate-level phosphorylation)
- 2 NADH (fed to ETC later)
- 2 pyruvate (3-carbon molecules)

"Glyco" = sugar, "lysis" = split. Literally: sugar-splitting.

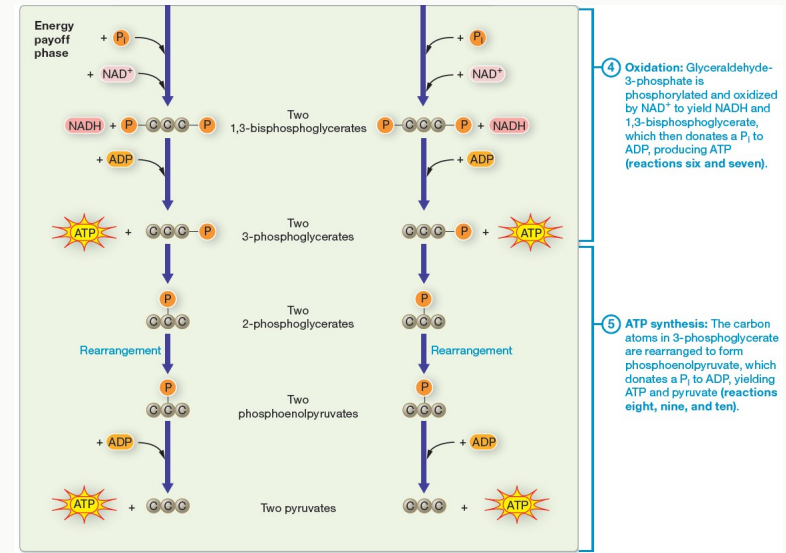


Fig. 23.6 — Energy payoff phase of glycolysis (steps 4–10)

Two fates for pyruvate: O₂ or no O₂?

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CONCEPT

O₂ present → AEROBIC

Pyruvate enters mitochondrion → becomes acetyl-CoA → citric acid cycle.

Total ATP yield: ~30 per glucose

Slow, clean, efficient. The default mode.

O₂ absent → ANAEROBIC

Pyruvate → lactate (cytosol). Regenerates NAD⁺ to keep glycolysis running.

Total ATP yield: only 2 per glucose

Fast, dirty, emergency. Sprinting muscle. Tumors (Warburg).

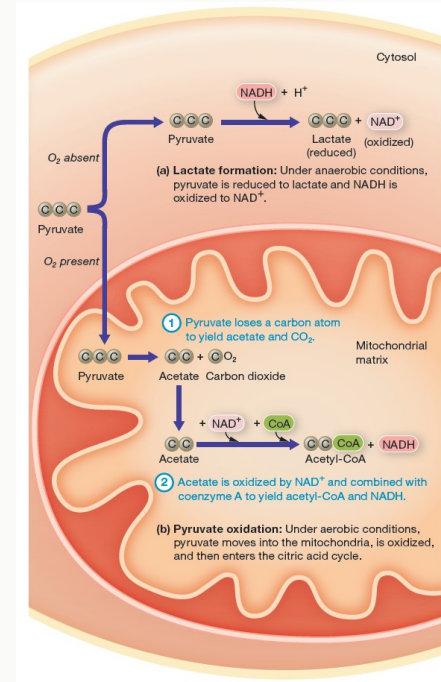


Fig. 23.9 — Pyruvate fates

Stage 2 — Citric Acid Cycle

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CONCEPT

AKA: Krebs cycle, TCA cycle

Where: mitochondrial matrix

Fuel: acetyl-CoA (from pyruvate, fatty acids, OR amino acids)

Per acetyl-CoA:

- 3 NADH
- 1 FADH₂
- 1 ATP
- 2 CO₂ (the CO₂ you exhale!)

Per glucose → 2 turns (2 pyruvate = 2 acetyl-CoA)

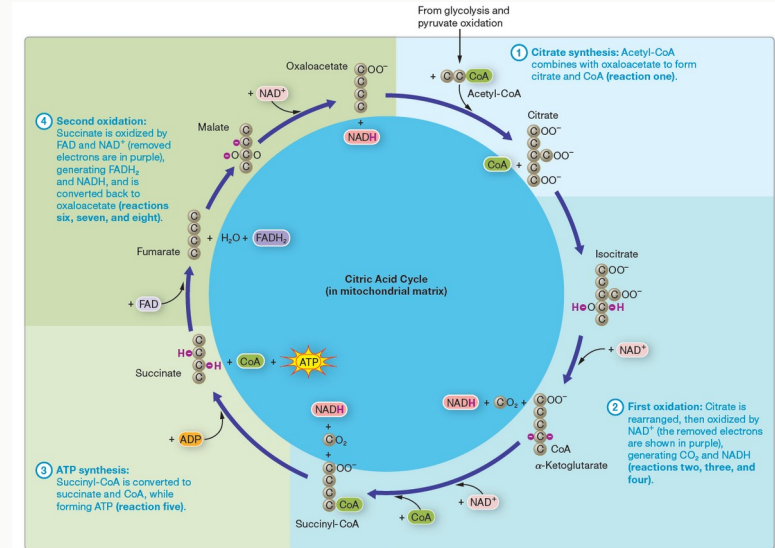


Fig. 23.10 — The citric acid cycle (one turn per acetyl-CoA)

Quick check: Where is the CO₂ coming from?

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CHECK

When you breathe out CO₂, where in the cell was that carbon originally?

- A** In the nucleus (from DNA)
- B** In the cytosol (from glycolysis)
- C** In the mitochondrial matrix (from the citric acid cycle)
- D** In the lungs (from the air you just inhaled)

Stage 3 — Oxidative Phosphorylation

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CONCEPT

This is where 90% of the ATP is made. Two linked machines: Electron Transport Chain + ATP Synthase.

ELECTRON TRANSPORT CHAIN

NADH and FADH_2 drop off electrons.
Electrons flow through 4 protein complexes in the inner membrane.
Energy is used to pump H^+ into the intermembrane space.

Final electron acceptor: $\text{O}_2 \rightarrow \text{H}_2\text{O}$

That's why you need oxygen. Without it, the chain stops, H^+ gradient collapses, ATP synthase stalls.

ATP SYNTHASE

A molecular turbine.

H^+ flows down its gradient through the rotor.
Rotation drives the phosphorylation of $\text{ADP} \rightarrow \text{ATP}$.

Yield: ~2.5 ATP per NADH, ~1.5 ATP per FADH_2

Chemiosmosis in action. Same principle your thylakoids use in photosynthesis (and mitochondria likely inherited from bacteria).

The proton pump + ATP synthase — visualized

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CONCEPT

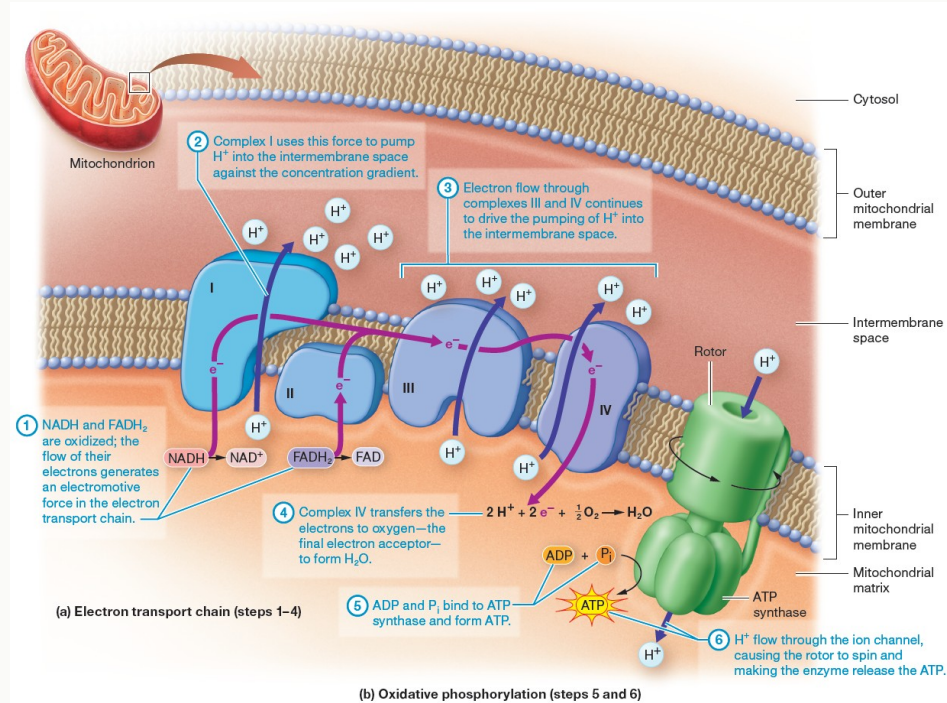


Fig. 23.11 — Electron transport chain (Complexes I–IV) pumps H^+ ; ATP synthase (green, right) lets H^+ flow back and makes ATP.

ATP yield: final tally per glucose

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CONCEPT

Stage	Location	ATP	NADH	FADH ₂
Glycolysis	Cytosol	+2	+2	0
Pyruvate → Acetyl-CoA	Matrix	0	+2	0
Citric acid cycle (×2)	Matrix	+2	+6	+2
Oxidative phosphorylation	Inner membrane	~26	—	—
TOTAL		~30–32 ATP		

Key insight: NADH and FADH₂ are just delivery trucks. The real ATP payoff is at the ETC.

Activity: Design the Poison

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ACTIVITY

The prompt: *You are an evil toxicologist. Where in the glucose catabolism pipeline would you attack to kill a cell the fastest? Pick a step, explain why, and predict what the patient would look like clinically.*

1 Think (2 min) On your own: pick a target. Which stage? Which enzyme?

2 Pair (3 min) Compare with neighbor. Different targets? Same?

3 Share (5 min) We'll map your poisons onto the pathway — and reveal real ones.

Real poisons that attack this pipeline

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CASE STUDY

Cyanide (CN^-) → **Blocks Complex IV of ETC**

Electrons can't reach O_2 . Proton gradient collapses. ATP synthesis stops in ~seconds. Seizures, death within minutes.

Carbon monoxide (CO) → **Binds hemoglobin + Complex IV**

Two hits: can't deliver O_2 AND can't use it. Cherry-red skin, confusion, collapse.

Rotenone (pesticide) → **Blocks Complex I**

Slows electron flow. Chronic low-dose exposure linked to Parkinson's.

Metformin (a DRUG, not a poison!) → **Partially inhibits Complex I**

Same mechanism — but gentle, controlled. Lowers liver glucose output. First-line diabetes drug.

MODULE 23.3

Fats and Proteins enter the same machine

All roads lead to acetyl-CoA

Three fuels, one destination

CONCEPT

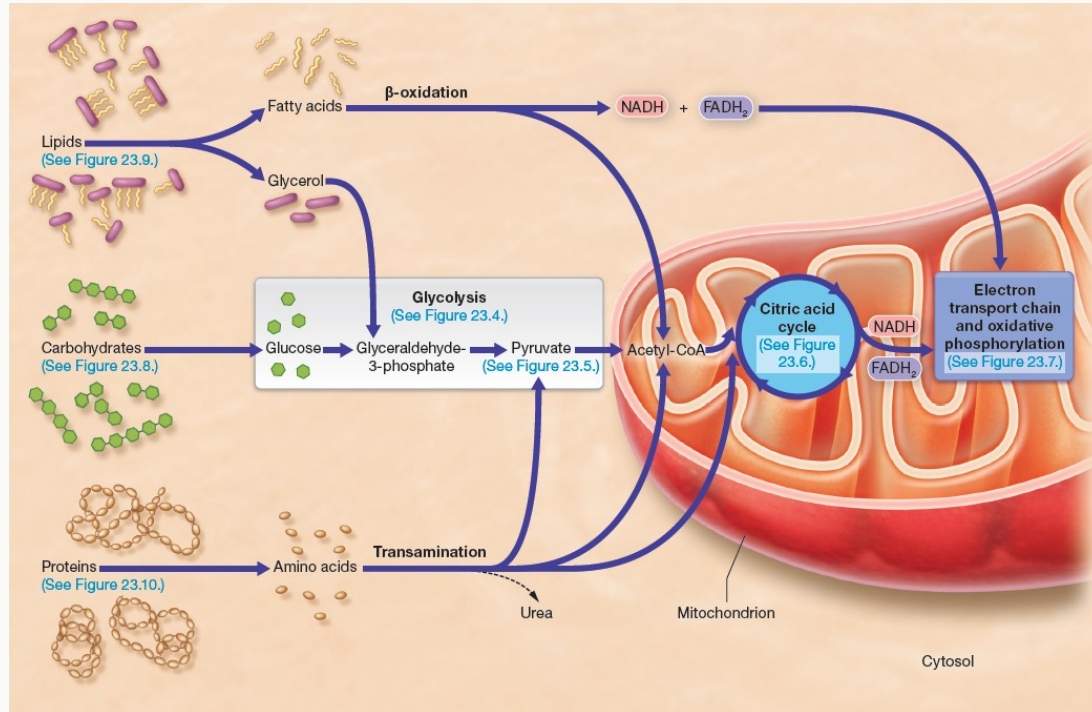


Fig. 23.13 — Fats (β -oxidation), glucose (glycolysis), and proteins (transamination/deamination) all converge on acetyl-CoA and the citric acid cycle.

Fatty acid catabolism (β -oxidation)

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CONCEPT

Fat: energy-dense storage

1 g fat = 9 kcal (vs 4 kcal for carbs/protein)

β -oxidation:

- Fatty acid enters mitochondrion
- Cleaved 2 carbons at a time
- Each cleavage $\rightarrow$ acetyl-CoA + NADH + FADH_2

A 16-C palmitate $\rightarrow$ ~106 ATP.

That's why fat is long-haul fuel (marathon, fasting, hibernation).

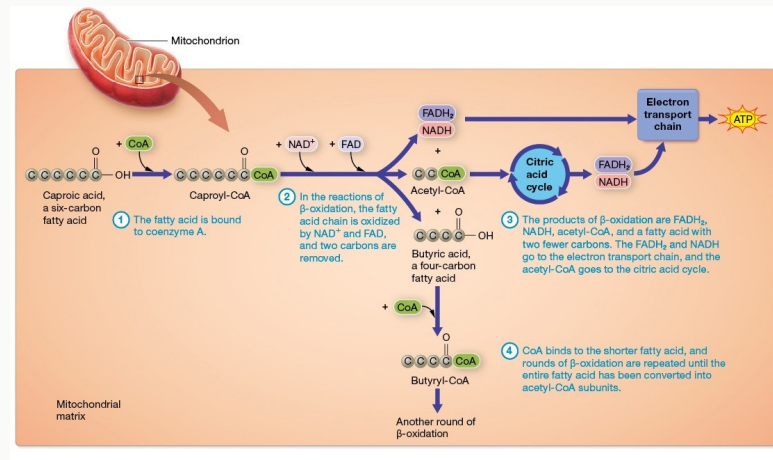


Fig. 23.14 — β -oxidation cuts 2 carbons per cycle

Amino acid catabolism

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CONCEPT

Two problems, two solutions

Problem 1: Amino acids have nitrogen. The citric acid cycle can't use it.

Solution: Strip the NH_2 off (deamination). Liver packages NH_3 as urea $\rightarrow$ urine.

Problem 2: 20 different amino acids = 20 different carbon skeletons.

Solution: Enzymes convert each into pyruvate, acetyl-CoA, or a citric acid cycle intermediate.

Burning protein is the body's LAST resort. Why? You're literally eating your own muscle.

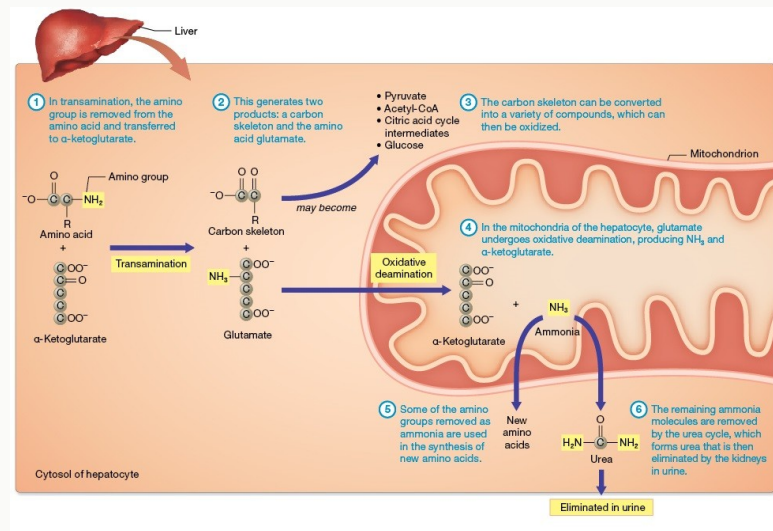


Fig. 23.15 — Transamination, oxidative deamination, and urea cycle

Brown Bag preview: matching patients to pathways

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CASE STUDY

This Saturday's RCS Quiz 2 — **six patients walk into Deana's Brown Bag Event.** *Each has a metabolic story we'll meet across this week.*

Steve

Module 23.2

Type 2 diabetes + metformin

*His insulin receptors are desensitized.
Metformin targets Complex I of ETC in liver.*

Mark

Module 23.3

High-protein / low-carb diet

What happens when you burn mostly fat? Ketones. His breath smells fruity.

Deana

Modules 23.1/23.6

Pharmacist, just had COVID

Recovering from infection = sky-high metabolic demand. Thyroid asked about it.

Clinical spotlight: Steve and metformin

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CASE STUDY

Meet Steve

58 y/o male, BMI 34

Recently diagnosed: Type 2 diabetes

New Rx: metformin 500 mg BID

Brown Bag complaint: *"My fasting glucose is coming down, but I feel tired and my stomach is upset. Am I just getting used to it?"*

What's happening at the mitochondrial level?

The mechanism

1. Metformin partially inhibits Complex I of ETC (primarily in liver cells).
2. Liver can't efficiently run oxidative phosphorylation.
3. Liver stops making new glucose (gluconeogenesis) → blood glucose drops.
4. Lactate rises slightly → fatigue. GI upset = gut microbiome shift.

Your job: *explain this to Steve in plain language.*

Think-pair-share: Why does Mark's breath smell fruity?

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ACTIVITY

Mark's case: 42 y/o on a low-carb ("keto") diet for 3 months. Lost 18 lbs. Wife says his breath smells "like nail polish remover" in the morning. He wants to know if this is normal.

Work it out:

1. Low carbs = liver runs out of oxaloacetate. What does that break in the citric acid cycle?
2. Acetyl-CoA piles up. Where does it go instead? (Hint: three molecules — acetone, acetoacetate, β -hydroxybutyrate.)
3. Which one is volatile and shows up in breath?

Concept check: trace the flow

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CHECK

A cell runs out of oxygen. Which of these will KEEP working (at least briefly)?

- A** Glycolysis only
- B** Citric acid cycle only
- C** Electron transport chain only
- D** All three — they're all aerobic

Answer: A — Glycolysis only

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CHECK



Glycolysis happens in the cytosol and does NOT require oxygen.

Why the other options fail:

B. Citric acid cycle: runs in the matrix, but it produces NADH/FADH₂ that need the ETC to unload. Without O₂ the ETC backs up, NAD⁺ can't be regenerated, and the cycle halts.

C. ETC: literally requires O₂ as the final electron acceptor. No O₂ = no flow = no ATP.

D: Glycolysis is anaerobic. That's why your sprinting muscles (briefly outrunning O₂ delivery) can still make ATP.

Clinical connection: *This is why tumors, which often have poor blood supply, love glycolysis (the Warburg effect).*

A&P review: what you already know from 40A

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A&P REVIEW

You've seen this before. What's new in 40C is the depth: mitochondrial structure, proton gradients, and how fat and protein plug into the same pipeline.

From 40A: cells use glucose for ATP

Now in 40C: three stages + substrate + O₂ requirements

From 40A: mitochondrion = powerhouse

Now in 40C: matrix vs. intermembrane space, cristae, ATP synthase

From 40A: we burn fats and carbs

Now in 40C: β -oxidation, amino acid deamination, ketogenesis

CONCEPT

Catabolism is a **CONVERGENT** pipeline.

Glucose, fat, and protein all funnel into the same machine: **acetyl-CoA → citric acid cycle → ETC → ATP.**

If you can...

- ✓ Draw the 3-stage pipeline from memory (glycolysis, CAC, OxPhos)
- ✓ Explain why ATP synthase needs a proton gradient
- ✓ Predict what cyanide/metformin/low-oxygen does to a cell

...you've nailed the backbone of this week.

Wednesday: Building, storing, regulating

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CONCEPT

Class 6 (Wed Apr 22): Anabolism + metabolic states + thermoregulation + nutrition. Juan's celiac story returns. Crystal's gallstone case launches.

23.4 **Anabolism** How do cells BUILD glycogen, fat, and protein? (Reverse of today.)

23.5 **Fed vs. fasted** Insulin and glucagon — the molecular switch that flips every few hours.

23.6 **Metabolic rate** Thyroid hormone, BMR, thermoregulation — why hypothyroid patients feel cold.

23.7 **Nutrition** Protein quality, lipoproteins, celiac → malabsorption revisited.

Between now and Wednesday

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CONCEPT

FOOTSIE (tutor AI) · Drill the 10 steps of glycolysis and the citric acid cycle. Today's class gave you the logic; now fill in the mechanism.

ARIA (practice AI) · Quiz yourself on ETC complexes, poison mechanisms, and ATP yield calculations.

Amerman Ch 23 · Read modules 23.4 (Anabolism) and 23.5 (Metabolic states). Skim 23.6 and 23.7.

Start RCS Quiz 2 · Open the Brown Bag Event case. Don't leave all six patients until Saturday night.

Due by Sat Apr 25, 11:59 PM · RCS Quiz 2 + Application Activity 2 (Crystal's gallstone case).

Questions?

See you Wednesday — we flip from breaking down to building up.

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